

Supplementary Tables 1 – 3

Supplementary Table 1. Distribution of hub genes and miRNAs across all six tissue types

Network	Hubs genes	Hub miRNAs	Methylated	Common Hubs (Genes)	Common Hubs (miRNAs)	Common Hubs (Methylated)
Basal	471	146	127	388	100	7 (<i>RUNX3</i> , <i>RORC</i> , <i>SPDEF</i> , <i>ELF5</i> , <i>SP11</i> , <i>HOXB2</i> , <i>STAT5A</i>)
Her2	449	131	81			
LumA	443	141	225			
LumB	440	141	161			
NormL	441	117	23			
TANT	445	128	113			

Supplementary Table 2. Distribution of bottleneck genes and miRNAs across all six tissue types

Network	Bottleneck genes	Bottleneck miRNAs	Methylated	Common Bottlenecks (Genes)	Common Bottlenecks (miRNAs)	Common Bottlenecks (Methylated)
Basal	335	19	5	145	3 (hsa-miR-34a, hsa-miR-149, hsa-miR-141)	3 (<i>SP11</i> , <i>SPDEF</i> , <i>RUNX3</i>)
Her2	341	16	6			
LumA	323	21	5			
LumB	317	13	5			
NormL	324	23	5			
TANT	302	20	6			

Supplementary Table 3. Distribution of hub and bottleneck lncRNAs across all six tissue types

Network	Hubs	Common Hubs	Bottlenecks	Common Bottlenecks
Basal	23 (MEG3, HCP5, SNHG3, KCNQ1OT1, UCA1, CDKN2B-AS1, HOTAIR, PVT1, GAS5, CRNDE, H19, XIST, DANCER, MALAT1, HOXA11-AS, SNHG1, AFAP1-AS1, SNHG6, NEAT1, CYTOR, DIRC1, CT62, and CLLU1)	15 (MEG3, HCP5, UCA1, CDKN2B-AS1, HOTAIR, PVT1, GAS5, H19, XIST, DANCER, MALAT1, SNHG1, AFAP1-AS1, NEAT1, and CLLU1)	7 (MEG3, PART1, HCP5, SNHG3, KCNQ1OT1, SNHG4, and NBR2)	1 (PART1)
Her2	21 (PVT1, MEG3, HCP5, H19, CDKN2B-AS1, GAS5, CRNDE, XIST, UCA1, DANCER, MALAT1, HOXA11-AS, HOTAIR, ZFAS1, CYTOR, AFAP1-AS1, NEAT1, SNHG1, OIP5-AS1, CT62, and CLLU1)		7 (PVT1, MEG3, PART1, HCP5, KCNQ1OT1, SNHG4, and NBR2)	
LumA	21 (H19, PVT1, MEG3, CASC2, SNHG1, HCP5, HOXA11-AS, HOTAIR, CRNDE, XIST, UCA1, DANCER, CYTOR, MALAT1, CDKN2B-		12 (H19, PVT1, MEG3, CASC2, ATXN8OS, SNHG1, SNHG3, KCNQ1OT1, PART1, HCP5, NBR2, and MIR17HG)	

LumB	AS1, GAS5, AFAP1-AS1, NEAT1, DIRC1, CT62, and CLLU1) 21 (MEG3, KCNQ1OT1, HCP5, H19, HOTAIR, PVT1, CRNDE, XIST, UCA1, DANCER, CDKN2B-AS1, SNHG12, SNHG1, AFAP1-AS1, MALAT1, CASC2, GAS5, NEAT1, DIRC1, CT62, and CLLU1)	7 (MEG3, ATXN8OS, PART1, KCNQ1OT1, SNHG4, HCP5, and NBR2)
NormL	18 (H19, CDKN2B-AS1, HOXA11-AS, HOTAIR, GAS5, CRNDE, PVT1, MALAT1, MEG3, UCA1, CYTOR, AFAP1-AS1, DANCER, XIST, NEAT1, SNHG1, HCP5, and CLLU1)	3 (H19, PART1, and BCYRN1)
TANT	21 (PVT1, H19, SNHG1, HCP5, NEAT1, CDKN2B-AS1, HOXA11-AS, HOTAIR, XIST, UCA1, DANCER, MALAT1, MEG3, GAS5, CYTOR, AFAP1-AS1, MIAT, SNHG16, KCNQ1OT1, CLLU1, and PLAC4)	7 (PVT1, H19, SNHG1, HCP5, PART1, NBR2, and EMX2OS)